Amino Acid

BY

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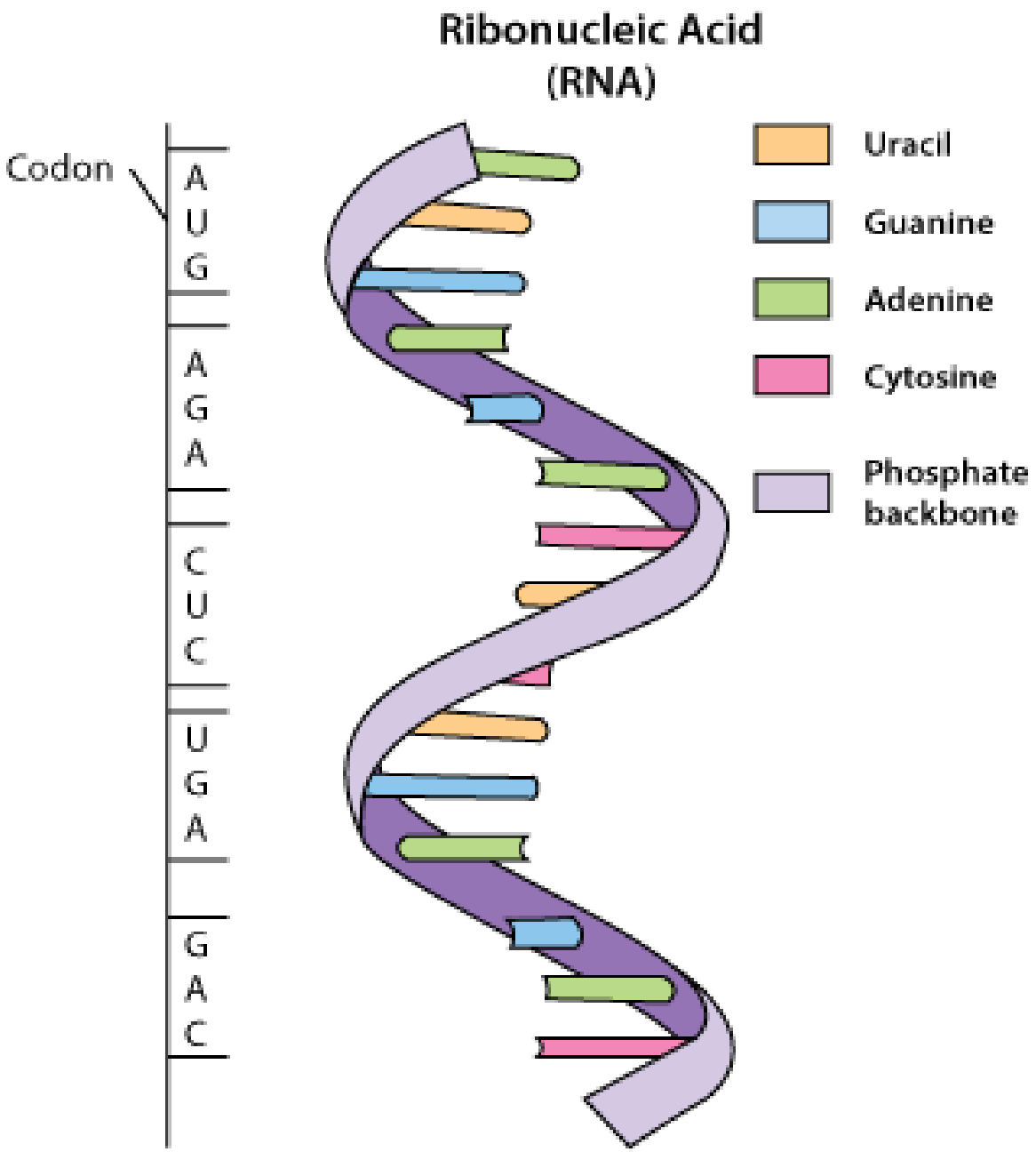
The Genetic Code

Marshall Nirenberg identified the codons that specify each amino acid.

RNA molecules of only 1 nucleotide and of specific 3-base sequences were used to determine the amino acid encoded by each codon.

The amino acids encoded by all 64 possible codons were determined.

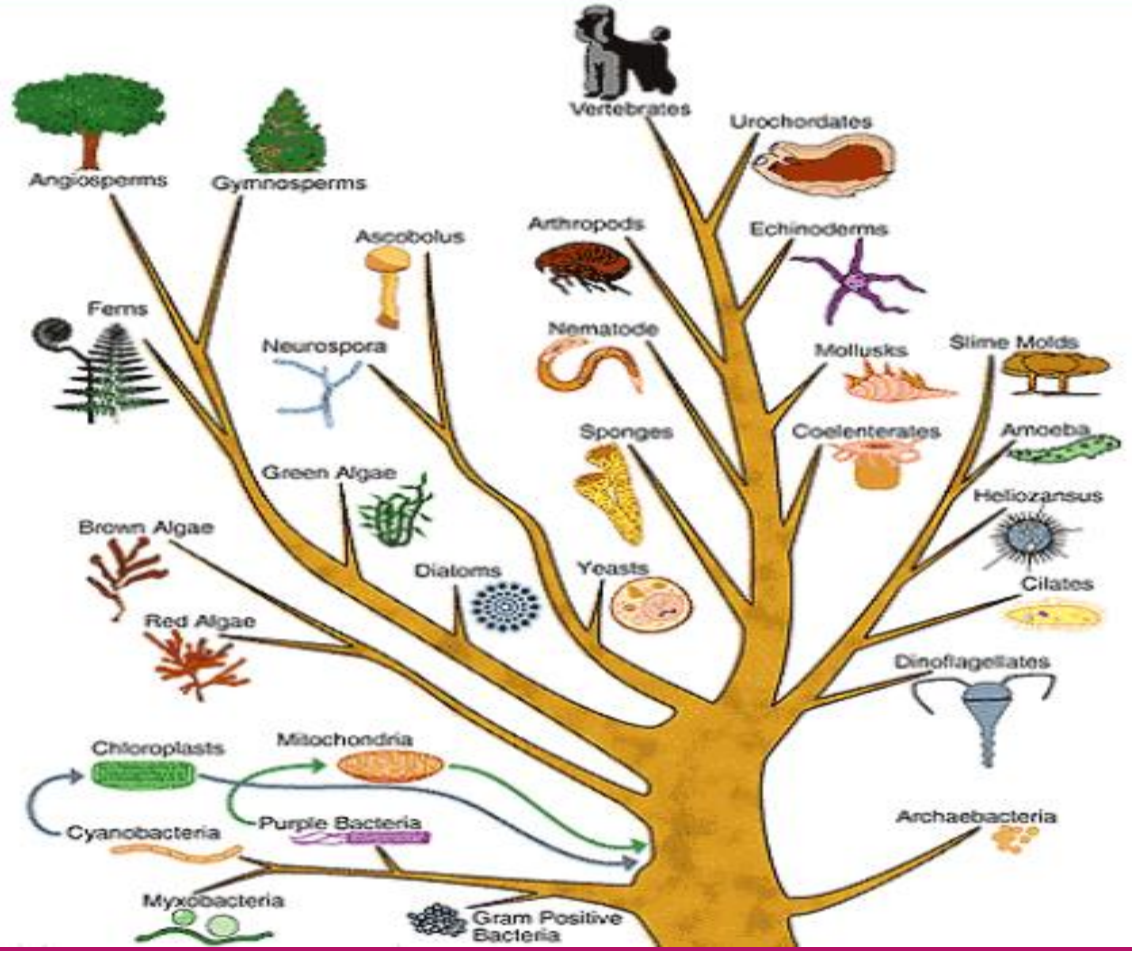


First letter	Second letter: U	Second letter: C	Second letter: A	Second letter: G	Third letter
U	UUU, UUC: Phe UUA, UUG: Leu	UCU, UCC, UCA, UCG: Ser	UAU, UAC: Tyr UAA: Stop UAG: Stop	UGU, UGC: Cys UGA: Stop UGG: Trp	U C A G
C	CUU, CUC, CUA, CUG: Leu	CCU, CCC, CCA, CCG: Pro	CAU, CAC: His CAA, CAG: Gln	CGU, CGC, CGA, CGG: Arg	U C A G
A	AUU, AUC, AUA: Ile AUG: Met	ACU, ACC, ACA, ACG: Thr	AAU, AAC: Asn AAA, AAG: Lys	AGU, AGC: Ser AGA, AGG: Arg	U C A G
G	GUU, GUC, GUA, GUG: Val	GCU, GCC, GCA, GCG: Ala	GAU, GAC: Asp GAA, GAG: Glu	GGU, GGC, GGA, GGG: Gly	U C A G

Amino acid names:

Ala = alanine
Arg = arginine
Asn = asparagine
Asp = aspartate
Cys = cysteine
Gln = glutamine
Glu = glutamate
Gly = glycine
His = histidine
Ile = Isolevcine
Leu = leucine
Lys = lysine
Met = methionine
Phe = phenylalanine
Pro = proline
Ser = serine
Thr = threonine
Trp = tryptophan
Tyr = Tyrosine
Val = valine

Phylogenetic tree construction



Phylogenetic tree?

- A tree represents graphical relation between organisms, species, or genomic sequence
- In Bioinformatics, it's based on genomic sequence

What do they represent?

- Root: origin of evolution
- Leaves: current organisms, species, or genomic sequence
- Branches: relationship between organisms, species, or genomic sequence
- Branch length: evolutionary time

Distance Matrix methods

- Calculate all the distance between leaves (taxa)
- Based on the distance, construct a tree
- Good for continuous characters
- Not very accurate
- Fastest method
  - UPGMA
  - Neighbor-joining

UPGMA

- Abbreviation of “Unweighted Pair Group Method with Arithmetic Mean”
- Originally developed for numeric taxonomy in 1958 by Sokal and Michener
- Simplest algorithm for tree construction, so it's fast!

How to construct a tree with UPGMA?

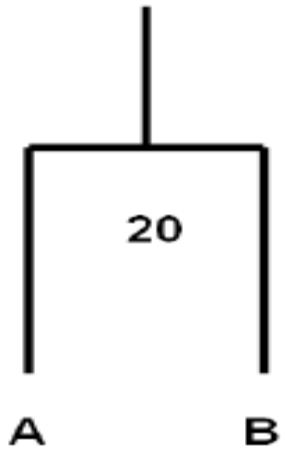
- Prepare a distance matrix
- Repeat step 1 and step 2 until there are only two clusters
- <u>Step 1:</u>

  Cluster a pair of leaves (taxa) by shortest distance
- <u>Step 2:</u>

  Recalculate a new average distance with the new cluster and other taxa, and make a new distance matrix

Example of UPGMA

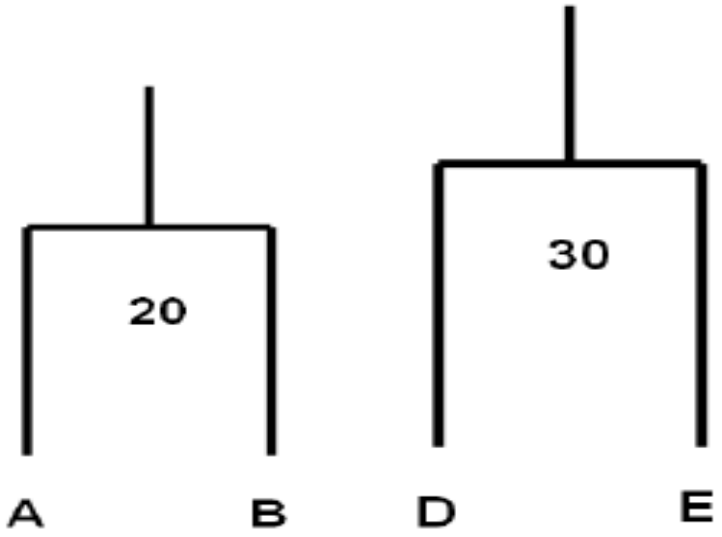
	A	B	C	D	E
A	0				
B	20	0			
C	60	50	0		
D	100	90	40	0	
E	90	80	50	30	0



- New average distance between AB and C is:
  - C to AB = (60 + 50) / 2 = 55
- Distance between D to AB is:
  - D to AB = (100 + 90) / 2 = 95
- Distance between E to AB is:
  - E to AB = (90 + 80) / 2 = 85

Example of UPGMA cont 1

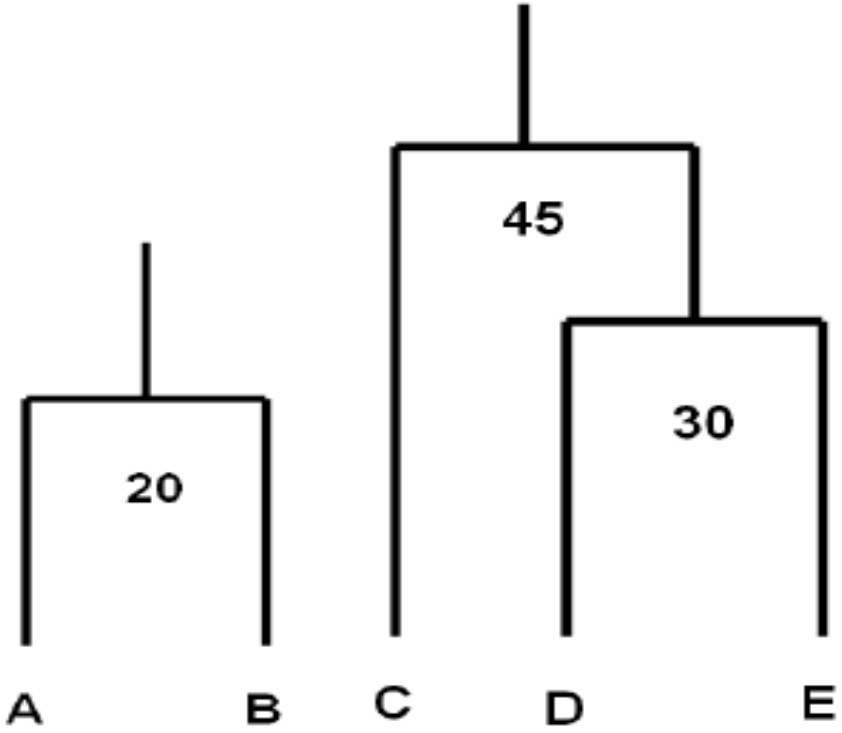
	AB	C	D	E
AB	0			
C	55	0		
D	95	40	0	
E	85	50	30	0



- New average distance between AB and DE is:
  - AB to DE = (95 + 85) / 2 = 90

Example of UPGMA cont 2

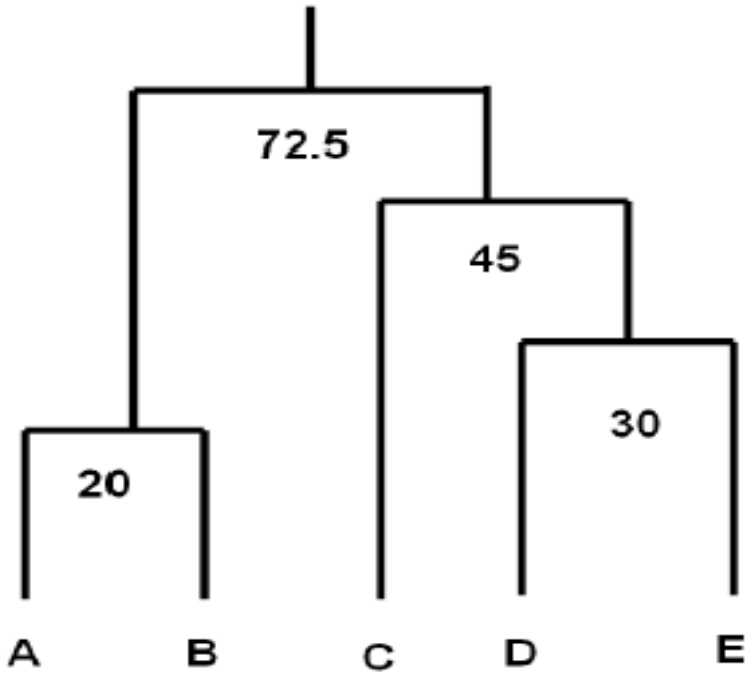
	AB	C	DE
AB	0		
C	55	0	
DE	90	45	0



- New Average distance between CDE and AB is:
  - CDE to AB = (90 + 55) / 2 = 72.5

Example of UPGMA cont 3

	AB	CDE
AB	0	
CDE	72.5	0



- There are only two clusters. so this completes the calculation!

Neighbor-joining method

- Developed in 1987 by Saitou and Nei
- Works in a similar fashion to UPGMA
- Still fast – works great for large dataset
- Doesn't require the data to be ultrametric
- Great for largely varying evolutionary rates

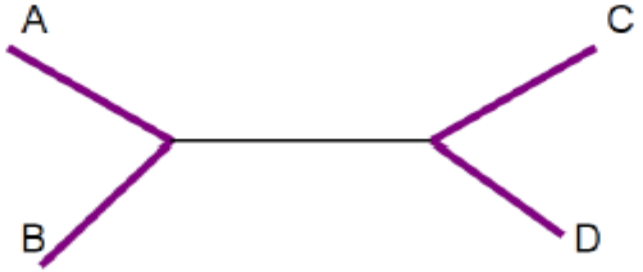
Neighbor Joining Algorithm

The Neighbor Joining method is a method for re-constructing phylogenetic trees, and computing the lengths of the branches of this tree.

In each stage, the two nearest nodes of the tree are chosen and defined as neighbors in our tree. This is done recursively until all of the nodes are paired together.

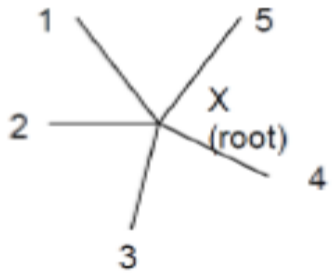
Neighbors are defined as a pair of OTU's (OTU=operational taxonomic units, or in other words – **leaves of the tree**), **who have one node connecting them**.

For instance, in the tree in figure 1, nodes A and B are neighbors (connected by only one internal node), and nodes C and D are neighbors, whereas nodes A and C (for example) are not neighbors.



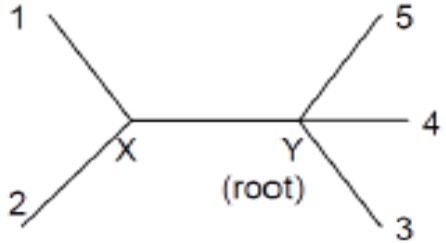
Neighbor Joining Algorithm
How do we find neighbors, and how de we construct our tree?

1. We start with a **star tree**:



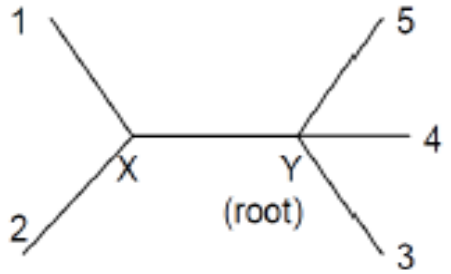
2. We define some kind of **distance** parameter between our nodes (1 through 5), and enter this parameter into a **distance matrix**. (The columns and rows of the matrix represent nodes, and the value i,j of the matrix represent the distance between node i and node j. Note that the matrix is symmetric, and that the diagonal is irrelevant, therefore only the top half (or lower half) are enough.)

3. **We pick the two nodes with the lowest value in the matrix** defined in step 2. These are **defined as neighbors.** For example, assuming nodes 1 and 2 are the nearest, we define them as neighbors

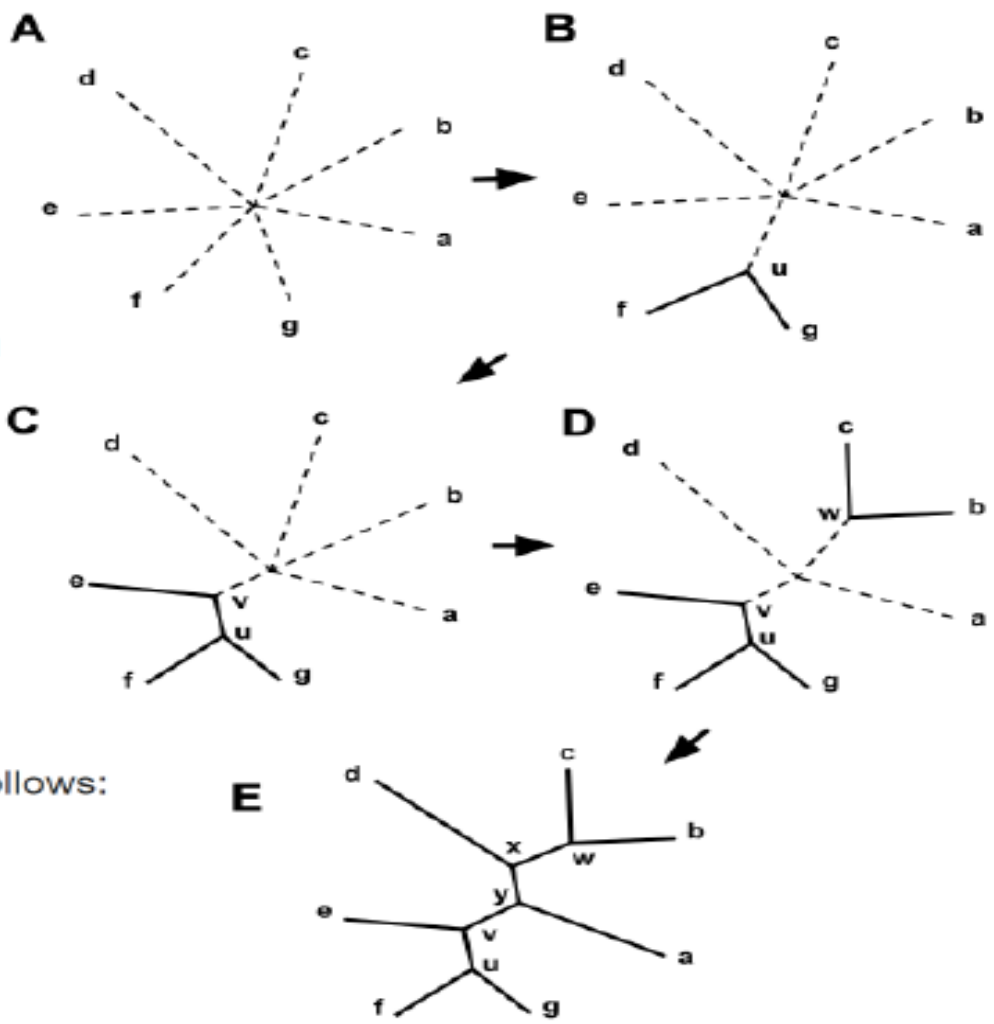


Neighbor Joining Algorithm

4. The new node we have added is defined as node X.
5. We now define the distance between node X and the rest of the nodes, and enter these distances into our distance matrix. We remove nodes 1 and 2 from our distance matrix.
6. We compute the branch lengths for the branches that have been joined (for figure 2(b), these are branches 1-X and 2-X) .
7. We repeat the process from stage 2 – once again we look for the 2 nearest nodes, and so on.



Starting with a star tree (A), the Q matrix is calculated and used to choose a pair of nodes for joining, in this case f and g. These are joined to a newly created node, u, as shown in (B). The part of the tree shown as solid lines is now fixed and will not be changed in subsequent joining steps.

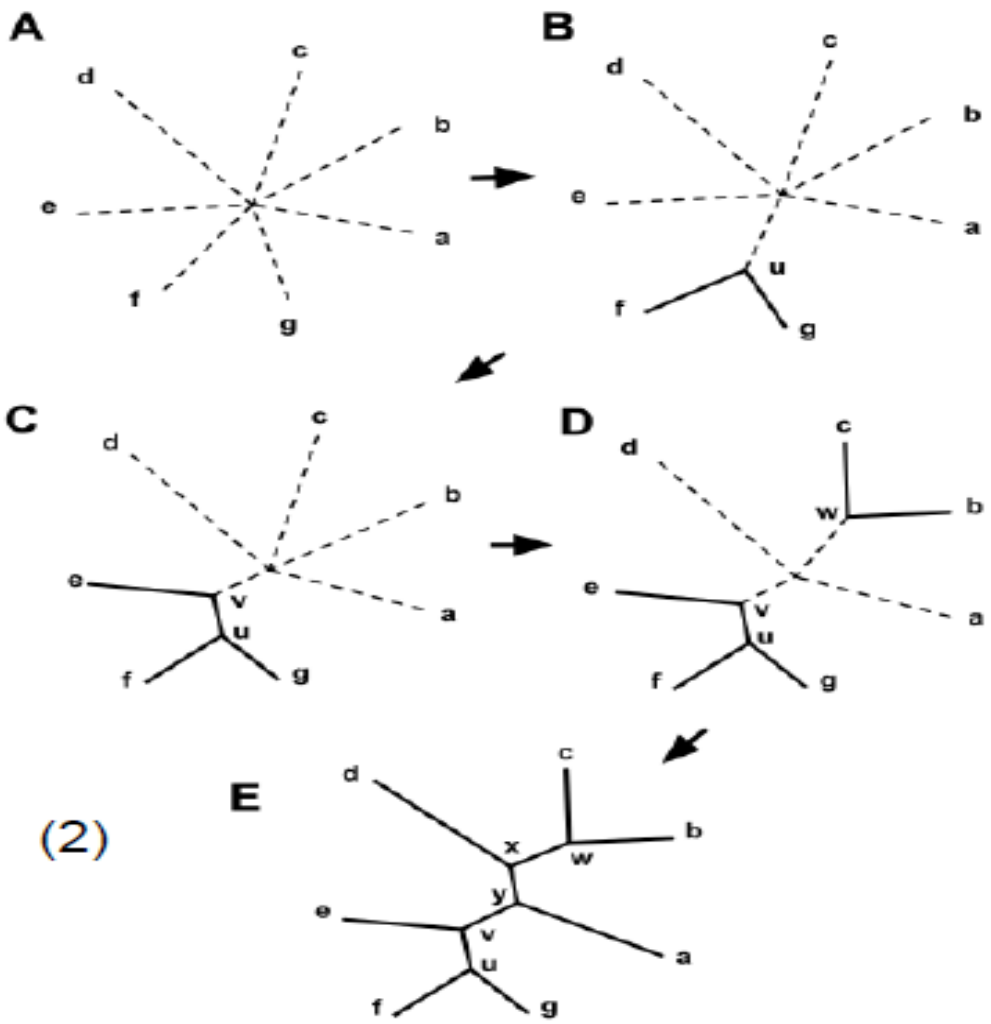


Based on a distance matrix relating the n taxa, calculate Q as follows:

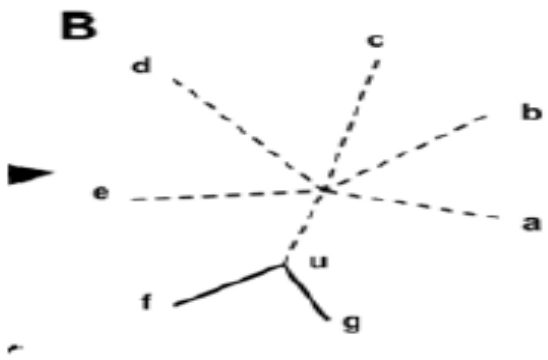
$$Q(i,j) = (n-2)d(i,j) - \sum_{k=1}^{n} d(i,k) - \sum_{k=1}^{n} d(j,k)$$

where $d(i,j)$ is the distance between taxa i and j.

The distances from node u to the nodes a-e are computed from equation (1). This process is then repeated, using a matrix of just the distances between the nodes, a,b,c,d,e, and u, and a Q matrix derived from it. In this case u and e are joined to the newly created v, as shown in (C). Two more iterations lead first to (D), and then to (E), at which point the algorithm is done, as the tree is fully resolved.



$$d(u,k) = \frac{1}{2}[d(f,k) + d(g,k) - d(f,g)] \quad (2)$$



Distance from the pair members to the new node

For each of the taxa in the pair being joined, use the following formula to calculate the distance to the new node:

$$\delta(f,u) = \frac{1}{2}d(f,g) + \frac{1}{2(n-2)}\left[\sum_{k=1}^{n} d(f,k) - \sum_{k=1}^{n} d(g,k)\right] \quad (2)$$

and:

$$\delta(g,u) = d(f,g) - \delta(f,u)$$

Taxa f and g are the paired taxa and u is the newly created node. The branches joining f and u and g and u, and their lengths, $\delta(f,u)$ and $\delta(g,u)$ are part of the tree which is gradually being created; they neither affect nor are affected by later neighbor-joining steps.

Closure